



7196 - How Extreme is Pegasus W? Testing Galaxy Evolution at the Lowest Stellar Masses

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Pegasus W Preimaging	NIRCam Imaging	(1) PEGW

Folder	Observation	Label	Observing Template	Science Target
	2	Scheduled PA 3 target sets Final new msa map	NIRSpec MultiObject Spectroscopy	(5) PEGW_MPT_GAIA

ABSTRACT

Ultrafaint dwarf galaxies (UFDs) are thought to be the fossils of galaxies that formed nearly all of their stars at early times, and were quenched by the bath of UV photons during the epoch of reionization. However, nearly all UFDs studied to date are satellites of the Milky Way or M31, where tidal and ram pressure stripping can significantly impact the history of a UFD. Recent high-resolution simulations suggest that in isolated UFDs, extended star formation will occur even in very low-mass halos, and reionization alone may not be responsible for quenching all UFDs.

Pegasus W is a newly discovered UFD on the far side of the Local Group. Based on HST imaging of the resolved stars, the galaxy was quenched long after reionization, ~ 7 Gyr ago. With a stellar mass of only 6.5×10^4 solar masses, Peg W may prove to be the most extreme example of low-mass, star forming galaxies discovered to date. The combination of Peg W's low stellar mass, extended star formation, and relative isolation means that matching its observed properties will provide a stringent test of galaxy formation models. However, understanding the nature of Peg W and its evolutionary history depends on measuring its dark matter (DM) content, for which we have no constraints.

We propose NIRSpec observations of the stars in Peg W to measure its DM halo mass to determine whether Peg W's total mass is large enough to have shielded itself from reionization. These observations will provide strong constraints on the efficacy of reionization in quenching the lowest mass galaxies, and more broadly test our understanding of the physics of galaxy formation and evolution in the lowest mass systems.

OBSERVING DESCRIPTION

Our goal is to obtain medium-resolution NIRSpec MSA spectra of a minimum of 35 Pegasus W stars in order to measure the stellar velocity dispersion, with which we will estimate the dark matter halo mass of Pegasus W.

NIRSpec MSA observations:

To achieve our target sample size, we will observe three target sets with exposures set to reliably recover velocities of stars 3 magnitudes fainter than the tip of the red giant branch. We will use the G140H/F100LP grism/filter setup that provides medium-resolution spectra in the 0.9-1.8 micron wavelength range. A setup with single shutter slits provides roughly 15 potential Pegasus W stars in one target set.

In five hours of exposure time (20 groups per integration, 2 integrations per exposure, and 6 dither positions, with the NRSIRS2 readout pattern), a

JWST Proposal 7196 (Created: Tuesday, June 9, 2026, 3:00:16PM Eastern Standard Time) - Overview

S/N over our wavelength range of 4 to 8 per pixel can be obtained for stars at our limiting magnitude. We will obtain 6 dither positions by designing 3 MSA configurations per target set and performing subpixel dithering at the planned position of each configuration to improve sampling of the line spread function.

NIRCam parallel observations:

We will observe NIRCam parallel observations in F140M and F277W. An astrometric analysis of the parallels will be used to determine the as-observed pointing and roll of the telescope, in order to attempt to improve upon corrections to the observed velocity based on the planned location of the sources in their shutters.

NIRCam pre-imaging observations:

The existing HST imaging will not provide adequate coverage of the MSA for performing target acquisition, due to the fact that all potential Pegasus W members will need to be placed on a single MSA quadrant, thereby substantially offsetting the MSA field of view from the HST imaging for any roll angle. Therefore, we require NIRCam preimaging observations. We will obtain a 3 row by 1 column mosaic with the FULL 3 dither pattern, which will provide the best spatial coverage for the MSA, regardless of assigned roll angle and quadrant on which the MPT finds the best success rate for observing Pegasus W members. We will observe with F115W and F277W, using 4 groups per integration, 2 integrations per exposure, with the BRIGHT1 readout pattern.

Proposal 7196 - Targets - How Extreme is Pegasus W? Testing Galaxy Evolution at the Lowest Stellar Masses

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	PEGW	RA: 23 53 14.9827 (358.3124279d) Dec: +22 06 8.68 (22.10241d) Equinox: J2000		
	Comments: Description=[]				
Fixed Targets	(5)	PEGW_MPT_GAIA	RA: 23 53 14.9955 (358.3124812d) Dec: +22 06 8.61 (22.10239d) Equinox: J2000		
	Comments: Description=[]				

Proposal 7196 - Observation 1 - How Extreme is Pegasus W? Testing Galaxy Evolution at the Lowest Stellar Masses

Observation	Proposal 7196, Observation 1: Pegasus W Preimaging									Tue Jun 09 20:00:16 GMT 2026
	Diagnostic Status: Warning									
Observing Template: NIRCam Imaging										
Diagnostics	(Pegasus W Preimaging (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.									
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
(Visit 1:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	PEGW	RA: 23 53 14.9827 (358.3124279d) Dec: +22 06 8.68 (22.10241d) Equinox: J2000							
	Comments: Description=[]									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module gap (large extended source)			
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order
	3	1	10.0		10.0		0.0		0.0	DEFAULT
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULL		3		STANDARD				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F115W	F277W	BRIGHT1	4	2	6	3	483.155	214066

Proposal 7196 - Observation 1 - How Extreme is Pegasus W? Testing Galaxy Evolution at the Lowest Stellar Masses

Special Requirements	Sequence Visits within 53.0 Days Visits Same PA 2 After 1 by 60.0 Days to <None specified>
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Proposal 7196 - Observation 2 - How Extreme is Pegasus W? Testing Galaxy Evolution at the Lowest Stellar Masses

Observation	Proposal 7196, Observation 2: Scheduled PA 3 target sets Final new msa map										Tue Jun 09 20:00:16 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Coordinated Parallel Template(s): NIRCcam Imaging										
Comments: We request these observations not be scheduled after observations of bright targets to avoid issues of persistence, given the faint nature of our targets.											
Diagnostics	(Visit 2:1) Warning (Form): Data Excess over lower threshold										
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:1) Warning (Form): The recommended value is 8 Reference Stars for this template.										
	(Visit 2:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets											
	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	PEGW_MPT_GAIA	RA: 23 53 14.9955 (358.3124812d) Dec: +22 06 8.61 (22.10239d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: F110W; Readout: NRSRAPID; 7 sources in 3 quads; [Reduced Accuracy]	SAME	F110W	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	NIRSpec MultiObject Spectroscopy					NIRCcam Imaging					
	TA Method: MSATA					Module: ALL					
	HFF Readout Mode: false					Subarray: FULL					
	Obtain Confirmation Images: No										
	Science Aperture: MSA Center										
	Primary Candidate List: ABOVE HB (87 sources)										
	Filler Candidate List: ABOVE HB (87 sources)										
Reference Stars	Spectral Overlap Map: jwst-nirspec-g140h										
	Spectral Overlap Threshold: 1.5										
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	402	358.310838	22.094217	20.18400001525879	1	423	358.316048	22.109758	21.111000061035156	
	1	412	358.301266	22.114767	20.726999282836914	1	427	358.352646	22.087704	21.420000076293945	
1	414	358.342723	22.129297	20.888999938964844	1	435	358.354789	22.097502	21.69700050354004		
1	420	358.322555	22.107316	21.062000274658203							
Dithers	#	Dither Type									
	1	NONE									

Proposal 7196 - Observation 2 - How Extreme is Pegasus W? Testing Galaxy Evolution at the Lowest Stellar Masses

	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G140H/F100LP)	c1		358.33132820833 333 Degrees 22.115918333333 333 Degrees	20.574296664910 076	0.125		1	2	2946.956
	2	1 (G140H/F100LP)	c1		358.33132820833 333 Degrees 22.115918333333 333 Degrees	20.574303701570 265	-0.125		1	2	2946.956
	3	1 (G140H/F100LP)	c2		358.33095283333 336 Degrees 22.116049166666 667 Degrees	20.574155938350 884	0.125		1	2	2946.956
	4	1 (G140H/F100LP)	c2		358.33095283333 336 Degrees 22.116049166666 667 Degrees	20.574162975063 842	-0.125		1	2	2946.956
	5	1 (G140H/F100LP)	c3		358.33095283333 336 Degrees 22.116049166666 667 Degrees	20.574155938350 884	0.125		1	2	2946.956
	6	1 (G140H/F100LP)	c3		358.33095283333 336 Degrees 22.116049166666 667 Degrees	20.574162975063 842	-0.125		1	2	2946.956
	7	1 (G140H/F100LP)	c4		358.33617025 Degrees 22.113641666666 67 Degrees	20.576110882027 706	0.125		1	2	2946.956
	8	1 (G140H/F100LP)	c4		358.33617025 Degrees 22.113641666666 67 Degrees	20.576117917798 875	-0.125		1	2	2946.956
	9	1 (G140H/F100LP)	c5		358.33579491666 666 Degrees 22.113772500000 003 Degrees	20.575970167278 097	0.125		1	2	2946.956
	10	1 (G140H/F100LP)	c5		358.33579491666 666 Degrees 22.113772500000 003 Degrees	20.575977203101 99	-0.125		1	2	2946.956
	11	1 (G140H/F100LP)	c6		358.33579491666 666 Degrees 22.113772500000 003 Degrees	20.575970167278 097	0.125		1	2	2946.956
	12	1 (G140H/F100LP)	c6		358.33579491666 666 Degrees 22.113772500000 003 Degrees	20.575977203101 99	-0.125		1	2	2946.956
	13	1 (G140H/F100LP)	c7		358.331186625 Degrees 22.112407222222 224 Degrees	20.574237123615 866	0.125		1	2	2946.956
	14	1 (G140H/F100LP)	c7		358.331186625 Degrees 22.112407222222 224 Degrees	20.574244159037 43	-0.125		1	2	2946.956

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	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	15	1 (G140H/F100LP)	c8		358.33081129166 67 Degrees 22.112537777777 778 Degrees	20.574096420489 66	0.125		1	2	2946.956
	16	1 (G140H/F100LP)	c8		358.33081129166 67 Degrees 22.112537777777 778 Degrees	20.574103455963 947	-0.125		1	2	2946.956
	17	1 (G140H/F100LP)	c9		358.33081129166 67 Degrees 22.112537777777 778 Degrees	20.574096420489 66	0.125		1	2	2946.956
	18	1 (G140H/F100LP)	c9		358.33081129166 67 Degrees 22.112537777777 778 Degrees	20.574103455963 947	-0.125		1	2	2946.956
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F140M	F277W	SHALLOW4	10	3	3	1	1599.779	214066	
	2	F140M	F277W	SHALLOW4	10	2	2	1	1062.94		
	3	F140M	F277W	SHALLOW4	10	3	3	1	1599.779		
	4	F140M	F277W	SHALLOW4	10	2	2	1	1062.94		
	5	F140M	F277W	SHALLOW4	10	3	3	1	1599.779		
	6	F140M	F277W	SHALLOW4	10	2	2	1	1062.94		
	7	F140M	F277W	SHALLOW4	10	3	3	1	1599.779		
	8	F140M	F277W	SHALLOW4	10	2	2	1	1062.94		
	9	F140M	F277W	SHALLOW4	10	3	3	1	1599.779		
	10	F140M	F277W	SHALLOW4	10	2	2	1	1062.94		
	11	F140M	F277W	SHALLOW4	10	2	2	1	1062.94		
	12	F140M	F277W	SHALLOW4	10	2	2	1	1062.94		
	13	F140M	F277W	SHALLOW4	10	3	3	1	1599.779		
	14	F140M	F277W	SHALLOW4	10	3	3	1	1599.779		
	15	F140M	F277W	SHALLOW4	10	3	3	1	1599.779		
	16	F140M	F277W	SHALLOW4	10	3	3	1	1599.779		
	17	F140M	F277W	SHALLOW4	10	3	3	1	1599.779		
	18	F140M	F277W	SHALLOW4	10	3	3	1	1599.779		
Special Requirements	No Parallel Attachments MSA Scheduled Aperture PA 20.5672 to 20.5672 Degrees (V3 241.99263 to 241.99263)										
	2 After 1 by 60.0 Days to <None specified>										